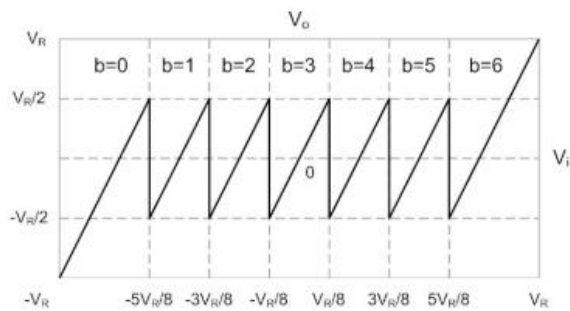


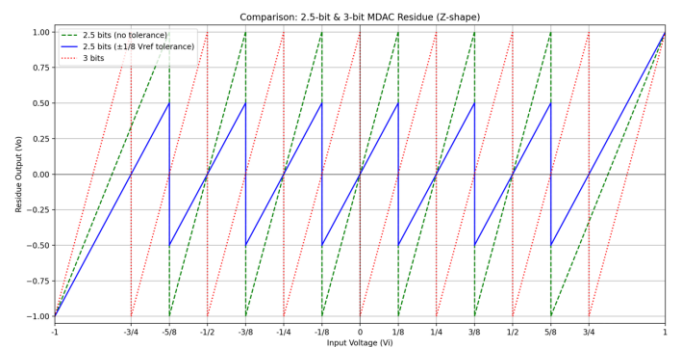
2.5-Bit RA Transfer Curve



Input Signal	Priority Encoder		
Vin (-1~1)	X2	X1	X0
0.625<Vin<0.875	1	1	0
0.375<Vin<0.625	1	0	1
0.125<Vin<0.375	1	0	0
-0.125<Vin<0.125	0	1	1
-0.125<Vin<-0.375	0	1	0
-0.375<Vin<-0.625	0	0	1
-0.625<Vin<-0.875	0	0	0

$$V_o = 4V_i - (V_a + V_b + V_c)$$

Vi	Thermo e6e5e4 e3e2e1	Enco de X2X1X0	DAC	Va Vb Vc	Vo
Vi<- 0.625	000000	000	-0.75	-1,-1,-1	4(vi+0.75)
-0.625 < Vi < -0.375	000001	001	-0.5	-1,-1,0	4(vi+0.5)
-0.375 < Vi < -0.125	000011	010	-0.25	-1,0,-0	4(vi+0.25)
-0.125 < Vi < 0.125	000111	011	0	0,0,0	4(vi)
0.125 < Vi < 0.375	001111	100	0.25	1,0,0	4(vi-0.25)
0.375 < Vi < 0.625	011111	101	0.5	1,1,0	4(vi-0.5)
Vi >0.625	111111	110	0.75	1,1,1	4(vi-0.75)



Blue (2.5-bit ADC): Gain=4, less comparators and can tolerate offset

Red (3-bit ADC): No matter if Gain=4 or 8 (Gain=8 than can't tolerate offset) still need more comparators

Green(2.5-bit ADC no tolerance): Gain=8, can't tolerate any offset (Vo will goes over Vref)